

Semi-total Domination in Unit Disk Graphs

Authors: Sasmita Rout, Gautam Kumar Das

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1. Problem Studied

- The paper studies the **Semi-total Dominating Set (STDS)** problem in Unit Disk Graphs (UDGs).
- A set $D_{t2} \subseteq V$ is a STDS if:
 1. D_{t2} is a dominating set.
 2. For every $u \in D_{t2}$, there exists $v \in D_{t2}$ such that $\text{dist}(u, v) \leq 2$.

2. Motivation and Applications

- STDS is a natural variant lying between classical dominating sets and total dominating sets.
- Relevant in wireless networks and sensor placements where both coverage and minimal connectivity are required.

3. Main Contributions

- Proved that the STDS problem is **NP-complete** in UDGs via a reduction from Vertex Cover on planar graphs.
- Proposed a **6-factor approximation algorithm** for STDS in UDGs.
- Introduced a $2 + \ln(D+1)$ approximation for general graphs, improving over the previous bound of $2 + 3\ln(D+1)$.
- Showed that a PTAS for STDS in UDGs exists with an approximation factor of $2 + \varepsilon$.

4. Techniques Used

- **NP-Completeness:** Used grid embedding and geometric gadget constructions to simulate vertex cover behavior.
- **Approximation Algorithm:**
 1. Compute a maximal independent set D .
 2. For each vertex in D lacking a neighbor within distance 2 in D , add a neighbor from $N(u)$ to set T .
 3. Return $D \cup T$ as the semi-total dominating set.

5. Results Summary

Graph Type	Approach	Approx. Factor	Time Complexity
UDG	Greedy + Local Fix	6	$O(nk)$
UDG (PTAS)	Local Search + Packing	$2 + \varepsilon$	$n^{O(1/\varepsilon \log \varepsilon)}$
General Graph	Set Cover-Based	$2 + \ln(D + 1)$	—

6. Comparison with Prior Work

- Improves over the $2 + 3 \ln(D + 1)$ result from Henning and Pandey (2019).
- The first work to give a dedicated algorithm and complexity result for STDS in UDGs.
- Relates to: $\gamma(G) \leq \gamma_{t2}(G) \leq \gamma_t(G)$.

7. Limitations / Assumptions

- Assumes no isolated vertices.
- Requires geometric embedding (disk centers must be known).
- Maximal independent set construction may not be efficient for very large or distributed settings.

8. Observations and Future Ideas

- Potential to explore distributed or dynamic versions of the 6-approximation algorithm.
- LP relaxation and rounding might yield tighter bounds.
- STDS in other geometric graphs or higher dimensions could be investigated.

(Independent) Roman Domination Parameterized by Distance to Cluster

Authors: Pradeesha Ashok, Gautam K. Das, Arti Pandey, Kaustav Paul, Subhabrata Paul

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1. Problem Studied

- The paper studies the Roman Domination and Independent Roman Domination problems.
- A Roman Dominating Function (RDF) is a function $f : V \rightarrow \{0, 1, 2\}$, such that every vertex v with $f(v) = 0$ has a neighbor u with $f(u) = 2$.
- An Independent RDF (IRDF) further requires that the set of vertices assigned 1 or 2 form an independent set.
- The study focuses on structural parameterization: specifically, the parameter is the size k of a minimum set of vertices whose removal results in a cluster graph (a disjoint union of cliques).

2. Motivation and Applications

- Roman Domination arises from defense strategies inspired by the Roman Empire.
- It models efficient resource placement (e.g., network monitoring, facility location) under constraints.
- Independent RDF ensures that no two deployed resources are adjacent — useful in scenarios requiring non-interference.

3. Main Contributions

- Proved that both the Roman Domination (RD-CVD) and Independent Roman Domination (IRD-CVD) problems are **FPT** when parameterized by distance to cluster.
- Gave algorithms with time complexity $4^k n^{\mathcal{O}(1)}$ for both problems.
- Proved matching **lower bounds** under the SETH assumption: the problems cannot be solved in $2^{\varepsilon k} n^{\mathcal{O}(1)}$ for any $\varepsilon < 1$, unless SETH fails.
- Proved that both problems do not admit a **polynomial kernel** unless $NP \subseteq coNP/poly$.

4. Techniques Used

- Used **structured branching** on cluster vertex deletion sets.
- Reduced the problem to a variant of **Set Cover with partition constraints**.
- Designed a dynamic programming algorithm with state space bounded by $2^{|S|} \cdot n^{\mathcal{O}(1)}$, where S is the cluster deletion set.
- Used reductions from the **d-Hitting Set** problem to prove lower bounds and kernelization hardness.

5. Results Summary

Problem	Parameter	Complexity	Remarks
RD-CVD	Cluster Deletion Size k	$4^k n^{\mathcal{O}(1)}$	FPT algorithm
IRD-CVD	Cluster Deletion Size k	$4^k n^{\mathcal{O}(1)}$	FPT algorithm
RD-CVD / IRD-CVD	Cluster Deletion Size k	No $2^{\varepsilon k} n^{\mathcal{O}(1)}$ algo	Unless SETH fails
RD-CVD / IRD-CVD	Cluster Deletion Size k	No Poly Kernel	Unless $NP \subseteq coNP/poly$

6. Comparison with Prior Work

- Extends the line of work on domination problems in parameterized complexity.
- Builds upon earlier results on classical Roman Domination and parameterization by treewidth and solution size.
- One of the first works to explore Independent Roman Domination with structural parameters.

7. Limitations and Assumptions

- The cluster vertex deletion set is assumed as part of the input.
- Results focus on deletion to cluster graphs; extension to other structural parameters like treewidth, neighborhood diversity is open.
- Real-world graphs may not always exhibit small cluster distance.

8. Observations and Future Directions

- Strongly bridges parameterized complexity and classical domination theory.
- Open directions:
 - Improve the time complexity (e.g., reduce base from 4).
 - Explore FPT status with parameters like modular-width, twin-cover.
 - Investigate kernelization under alternative parameters.

Total Roman Domination and Total Domination in Unit Disk Graphs

Authors: Sasmita Rout, Pawan Kumar Mishra, Gautam Kumar Das

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1. Problem Studied

- The paper focuses on two problems in Unit Disk Graphs (UDGs):
 1. The **Total Roman Dominating Set (TRDS)** problem.
 2. The **Total Dominating Set (TDS)** problem.
- A **Total Roman Dominating Function (TRDF)** is a Roman dominating function where the subgraph induced by non-zero valued vertices has no isolated vertex.
- The objective is to find a TRDF of minimum weight (sum of vertex values).

2. Motivation

- These problems have direct applications in wireless sensor and ad-hoc networks.
- Ensures both domination and resilience (total property), mimicking fault tolerance and mutual communication.
- UDGs model proximity-based communication (e.g., sensors with fixed radius).

3. Main Contributions

- Proposed a **10.5-approximation algorithm** for TRDS in UDGs with time complexity $\mathcal{O}(|V| \log k)$, where k is the number of vertices assigned Roman value 2.
- Improved approximation factor over previous best-known 12-approximation.
- Proposed a new **7.79-approximation algorithm** for the TDS problem in UDGs, improving over the previous 8-factor approximation.
- TDS algorithm runs in $\mathcal{O}(|V| + |E|)$ time.

4. Techniques Used

- For TRDS:
 - Construct maximal independent set V_2 (assigned value 2).
 - Add neighboring vertices V_1 (assigned value 1) to ensure no isolated vertices in $V_1 \cup V_2$.
- For TDS:
 - BFS to layer graph into sets S_i , construct independent sets I_i , and parent sets T_i for total property.

5. Results Summary

Problem	Approx. Factor	Time Complexity	Improvement
TRDS in UDGs	10.5	$\mathcal{O}(V \log k)$	Improves previous 12-factor
TDS in UDGs	7.79	$\mathcal{O}(V + E)$	Improves previous 8-factor

6. Comparison with Prior Work

- Builds on prior work by Shang et al. (RDS in UDGs) and Jena & Das (TDS in UDGs).
- Avoids distributed algorithm overhead from previous CRDS approximations.
- Provides improved constants for both TRDS and TDS.

7. Assumptions and Limitations

- Requires UDG embedding (disk centers in $\mathbb{R}^2$ known).
- Performance guarantees are worst-case; empirical performance not evaluated.
- TRDS approximation relies on geometric packing bounds (disk overlaps).

8. Observations and Future Work

- Provides structural lemmas and tight bounds on packing independent disks.
- Opens avenues for:
 - Distributed versions of the algorithms.
 - Extending to dynamic or weighted UDGs.
 - Empirical evaluation on real-world networks.